

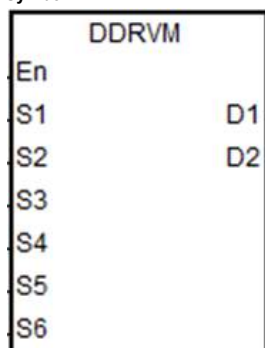
API	Instruction			Operand	Description
2709	D	DRVM		S ₁ , S ₂ , S ₃ , S ₄ , S ₅ , S ₆ , D ₁ , D ₂	Mark alignment positioning

Device	X	Y	M	S	T	C	HC	D	FR	SM	SR	E	K	16#	"\$"	F
S_1								●	●				○	○		
S_2								●	●				○	○		
S_3	○		○													
S_4								●	●				○	○		
S_5								●	●				○	○		
S_6								●	●				○	○		
D_1		○														
D_2		○	○													

Data type	BOOL	WORD	DWORD	LWORD	UINT	INT	DINT	LINT	REAL	LREAL	TMR	CNT	STRING
S_1			●				●						
S_2			●				●						
S_3	●												
S_4			●				●						
S_5			●				●						
S_6			●				●						
D_1	●												
D_2	●												

Pulse Instruction	16-bit instruction	32-bit instruction
—	—	AS

Symbol



- S_1 : Target number of output pulses
- S_2 : Target output frequency
- S_3 : Input point number for receiving the external interrupt signal
- S_4 : The number of output pulses at the end for the fore mask section
- S_5 : The number of output pulses at the beginning for the rear mask section
- S_6 : The number of output pulses in the ramp-down process after the interrupt signal is received
- D_1 : Pulse output device
- D_2 : Direction pulse output device

Explanation

- This instruction performs mark alignment positioning. When the value of S_1 is 0, it indicates the output is a maximum 32-bit value (with a sign) and the output does not perform the ramp-down stop until receiving the mark signal.
If the value of S_1 is not 0 and the external interrupt does not occur, the target number output pulses are output. The range of S_1 is between -2,147,483,648 to +2,147,483,647, and the +/- sign means the positive or negative direction.
- When the value of S_2 is less than or equal to 0, the output is not enabled. If the output frequency is greater than the maximum frequency, the PLC processes it as the maximum output frequency.
- S_3 can specify the M or X device as the input point. The input source signal is affected by the scan cycle if the selected input point is not in X0.0–X0.15 in the PLC.
- You must use the instruction by combining the input point number and the external interrupt program to achieve the real-time ramp-down output. For the rising-edge or falling-edge trigger for the external interrupt, select the triggers in the external interrupt program in ISPSOft.
 - Using the external interrupt (e.g. I0xx, I1xx): when the external interrupt occurs, the ramp-down stop is performed after the mark signal is received. No operation is performed if the external interrupt is not enabled.
 - Without using the external interrupt (I0xx, I1xx): When the instruction is executed, the ramp-down stop is performed after the mark signal is received if the rising edge occurs at X point or M device. It is affected by the scan time.
- The number of output pulses in the fore mask section is between 1– S_4 . When the setting value for S_4 is 0, it means the fore- mask section function is disabled. When the number of pulses to be masked exceeds the value of S_1 , the instruction takes the target number of output pulses S_1 as the number to be masked. If the external input trigger occurs within the number of output pulses to be masked, the external input interrupt is automatically invalid.

6. The number of output pulses in the rear mask section is between S_5 and S_1 . When $S_5 = 0$ or $S_5 \geq S_1$, the rear mask section function is disabled. When the external input trigger occurs in the masked sections, the external input interrupt is automatically invalid. When $S_4 > S_5$, it indicates the external input interrupt is invalid in the output process.
7. If the fore and rear mask sections are both set and $S_4 < S_5$, the valid input interrupt occurs in the section between $S_4 + 1$ and $S_5 - 1$.
8. For PLC with FW V1.04.00 (V1.04.00 included) or previous versions:

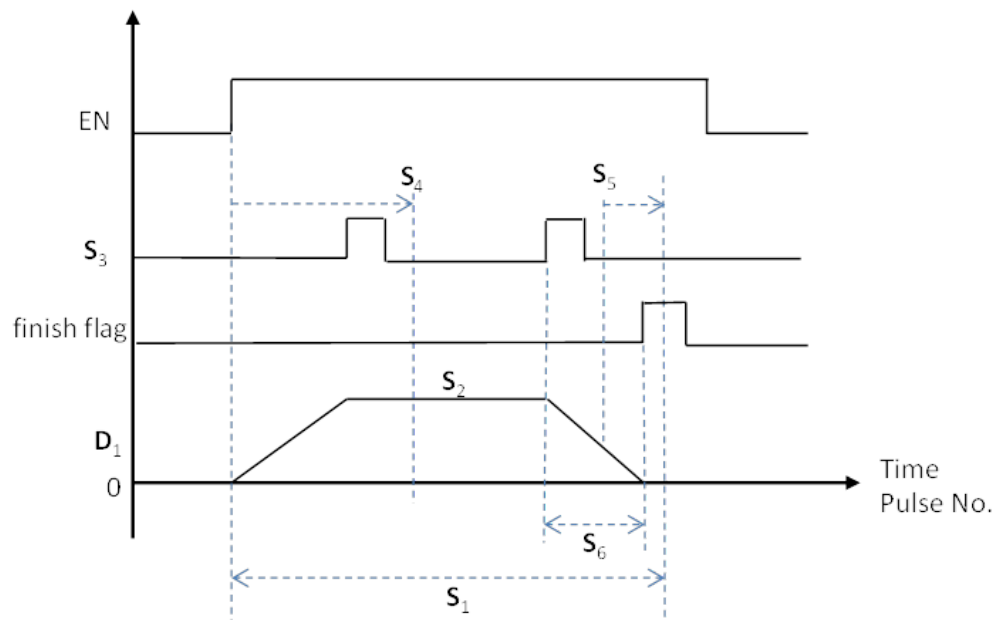
S_6 : the number of output pulses in the ramp-down	Description
If S_6 is set to 0 after the mark signal is received	the ramp-down stop is performed based on the ramp-down time
If $S_6 = -1$ or < 0 after the mark signal is received	the output immediately stops
If the setting value of S_6 is not enough to achieve the ramp-down stop within the ramp-down time,	The instruction limits the target frequency and performs the ramp-down stop in accordance with the set ramp-down time
If S_6 is greater than the number of pulses output within the ramp-down time,	The instruction outputs the number of redundant pulses with the same output frequency when the interrupt is triggered, and then performs the ramp-down process

9. For PLC with FW V1.06.00 (V1.06.00 included) or later: If S_6 is set to 0 after the mark signal is received, it means that the ramp-down stop is performed based on the ramp-down time. If S_6 is set to -1 after the mark signal is received, it means that the output immediately stops. If S_6 is set to > 0 after the mark signal is received and if the setting value of S_6 is not enough to achieve the ramp-down stop within the ramp-down time, the instruction limits the target frequency and performs the ramp-down stop in accordance with the set ramp-down time. If S_6 is greater than the number of pulses output within the ramp-down time, the instruction outputs the number of redundant pulses with the same output frequency when the interrupt is triggered, and then performs the ramp-down process.

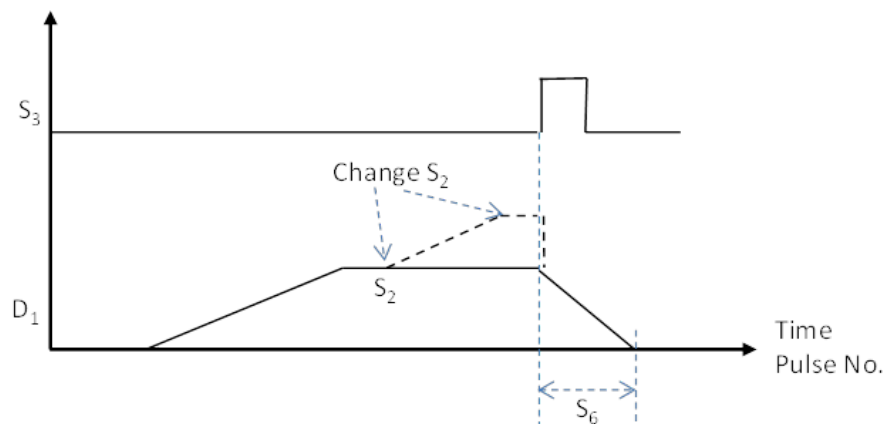
S_6 : the number of output pulses in the ramp-down	Description
If $S_6 = > 0$ after the mark signal is received	The instruction limits the target frequency and performs the ramp-down stop in accordance with the set ramp-down time
If the setting value of S_6 is not enough to achieve the ramp-down stop within the ramp-down time,	
If $S_6 = > 0$ after the mark signal is received	The instruction outputs the number of redundant pulses with the same output frequency when the interrupt is triggered, and then performs the ramp-down process
If S_6 is greater than the number of pulses output within the ramp-down time,	
If S_6 is set to 0 after the mark signal is received	The ramp-down stop is performed based on the ramp-down time
If $S_6 = -1$ after the mark signal is received	The output immediately stops
If $S_6 = < -1$ after the mark signal is received	The instruction does NOT limit the target frequency and after outputting the number of output pulses, the output stops immediately
If the setting value of S_6 is not enough to achieve the ramp-down stop within the ramp-down time,	
If $S_6 = < -1$ after the mark signal is received	The instruction outputs the number of redundant pulses with the same output frequency when the Interrupt is triggered, and then performs the ramp-down process
If S_6 is greater than the number of pulses output within the ramp-down time,	

- Added new SR652, SR653 for displaying number of pulse in ramp-up area and new SR654, SR655 for displaying number of pulse in ramp-down area.

10. If the output has entered the ramp-down process when the external input interrupt trigger occurs, the instruction completes the output of the number of pulses specified by S_6 .
11. See the following timing diagram about the output, interrupt trigger, mask and relevant flags. For the completion flag, refer to the Completion flag SM for the axes in DDRVI instruction (API 2706).

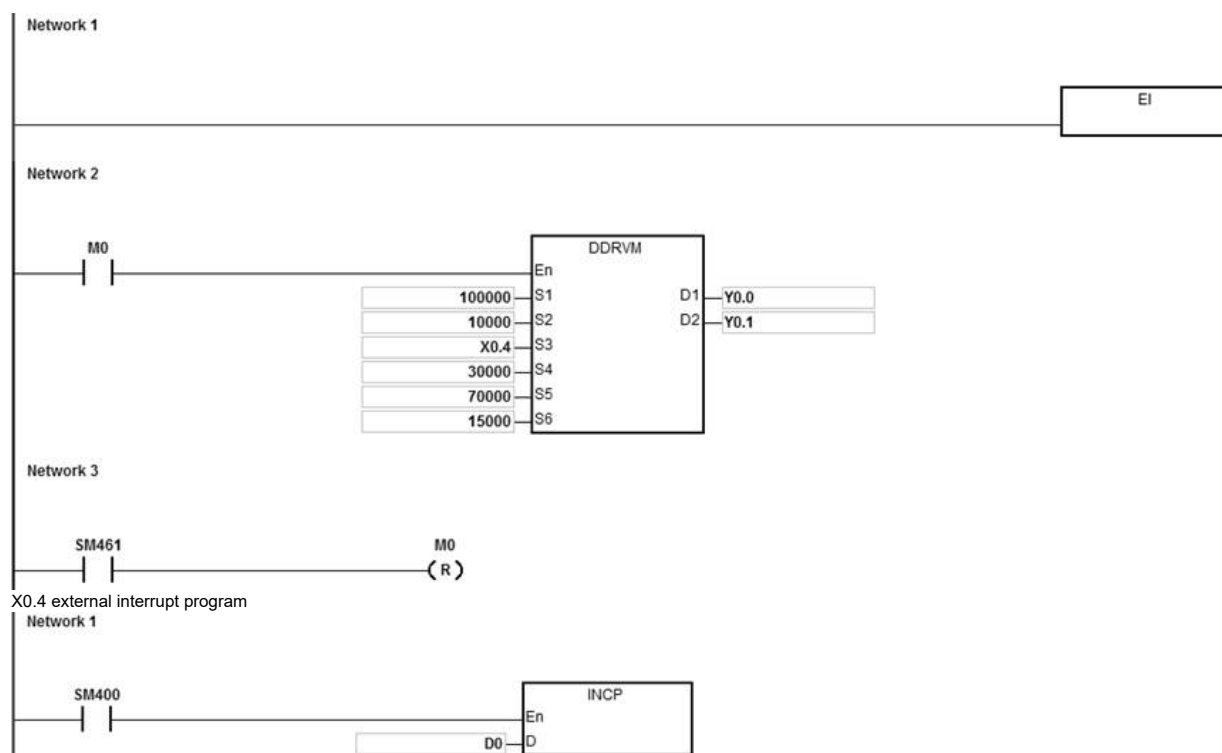


12. The target frequency **S₂** of the instruction can be modified in the process of the pulse output. Since the number of output pulses in the ramp-down section has been set when the instruction is executed, the timing diagram for changing the frequency is shown as the following dashed line.



13. Refer to the **DDRV1** instruction (API 2706) for the selection of pulse output devices for **D₁** and **D₂**. The output points for **D₁** can only be even-numbered such as Y0.0, Y0.2 or Y0.10. If the output points are not the recommended ones, the direction output **D₂** is affected by the scan cycle.
14. The instruction cannot be used in the ST programming language, interrupt tasks or function block which is called only once.

Example 1

**Notes**

- When M0 changes from OFF to ON, Y0.0 starts to output pulses. After outputting more than 30,000 pulses, the external interrupt is detected on X0.4. The value in D0 increases by 1 and the pulse output stops after 15,000 pulses ($S_6 = 15,000$). If the interrupt does not occur, the output does not stop until 100,000 pulses are output.
- If the number of output pulses is between 1–30,000, they are in the fore mask section. The external interrupt occurs at X0.4 at the moment and the instruction does not perform the ramp-down stop operation.
- If the number of output pulses is between 70,000–100,000, they are in the rear mask section. The external interrupt occurs at X0.4 at the moment and the instruction still does not perform the ramp-down stop operation.
- When the pulse output completes, SM461 is ON and M0 is reset.

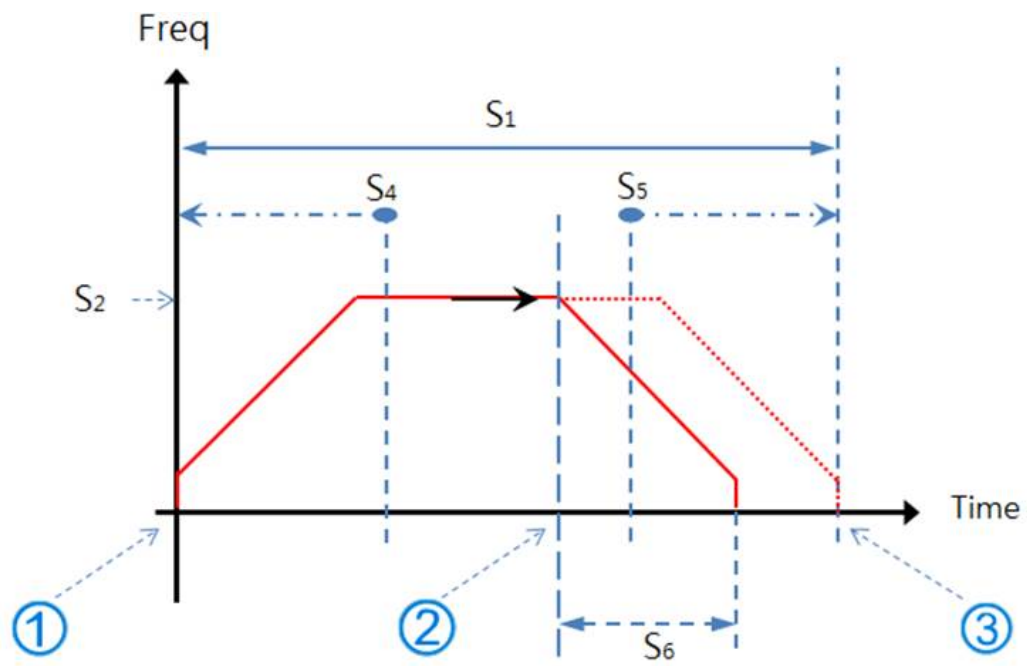
Additional remarks

- The alignment mark function and mask function:
 - When DDRVM is executed, the pulse output starts.
 - When the interrupt occurs for the mark alignment, the ramp-down process starts and the output stops after the number of the pulses specified by S_6 .
 - When the interrupt for the mark alignment does not occur or has no effect, DDRVM stops outputting the pulses after the target number of pulses specified by S_4 .

S_4 : The number of output pulses in the fore mask section. If the interrupt occurs in this section, the interrupt is ineffective for the mark alignment.

S_5 : The number of output pulses in the rear mask section. If the interrupt occurs in this section, the interrupt is ineffective for the mark alignment.

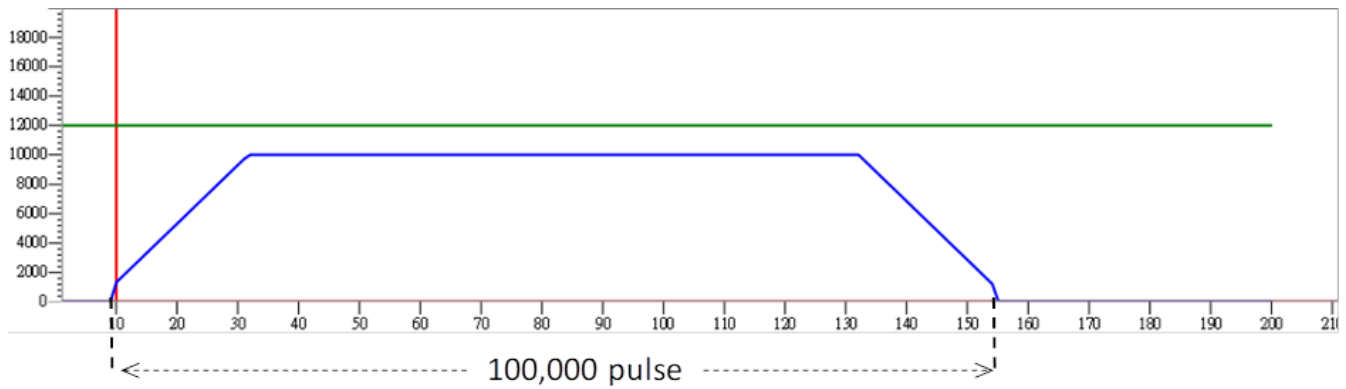
S_2 : The target output frequency



- The mark alignment function is enabled in the ramp-up and target frequency and ramp-down sections. (Horizontal coordinate (x axis) represents TIME and vertical coordinate (y axis) represents PULSE.)

When the mark alignment function is not enabled, set $S_1 = 100,000$; the output stops after 100,000 pulses.

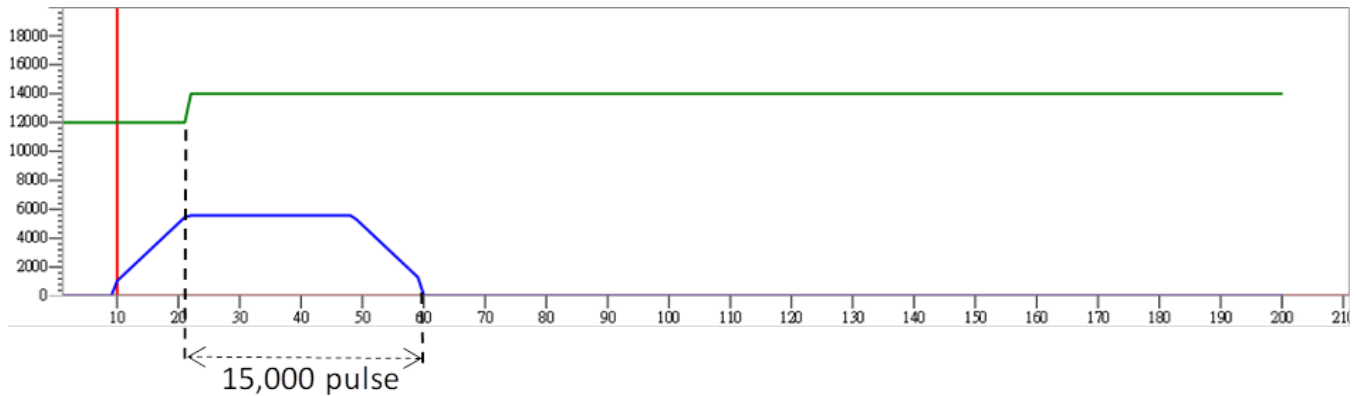
- Green line: signal of mark alignment
- Blue line: the actual output speed



Case 1: when the mark alignment occurs in the ramp-up section

- Green line: signal of mark alignment
- Blue line: the actual output speed

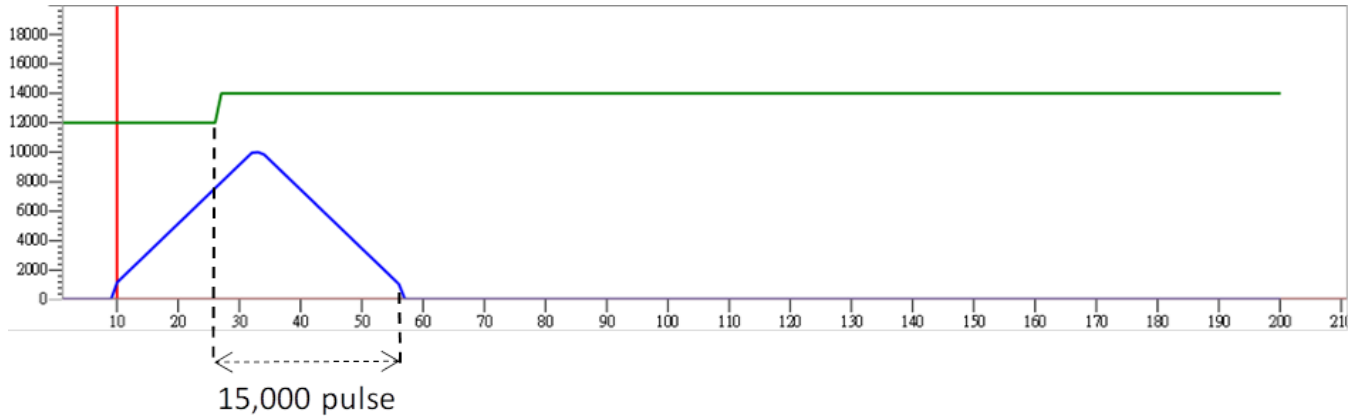
The operation of the mark alignment (green line: OFF -> ON) begins when the number of pulses is 3000. The target frequency of 10 kHz cannot be reached even if S_g is 15,000.



Case 2: when the mark alignment occurs in the ramp-up section

- Green line: signal of mark alignment
- Blue line: the actual output speed

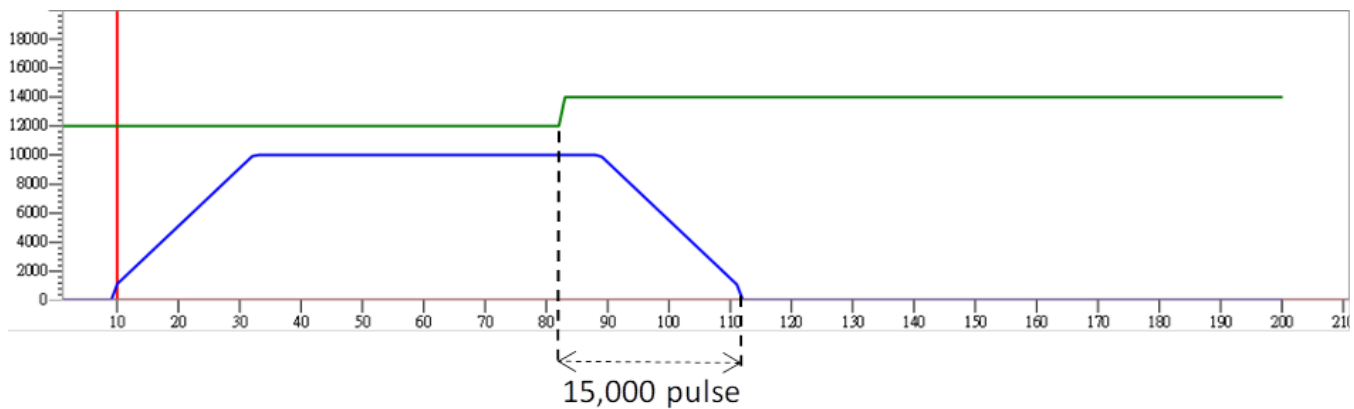
The operation of the mark alignment (green line: OFF -> ON) begins when the number of pulses is 6000. The target frequency of 10 kHz can be reached if S_6 is 15,000.



When the mark alignment occurs in the target frequency section

- Green line: signal of mark alignment
- Blue line: the actual output speed

The operation of the mark alignment (green line: OFF -> ON) begins when the number of pulses is 50,000. Set $S_6 = 15,000$; the output stops after 15,000 pulses.

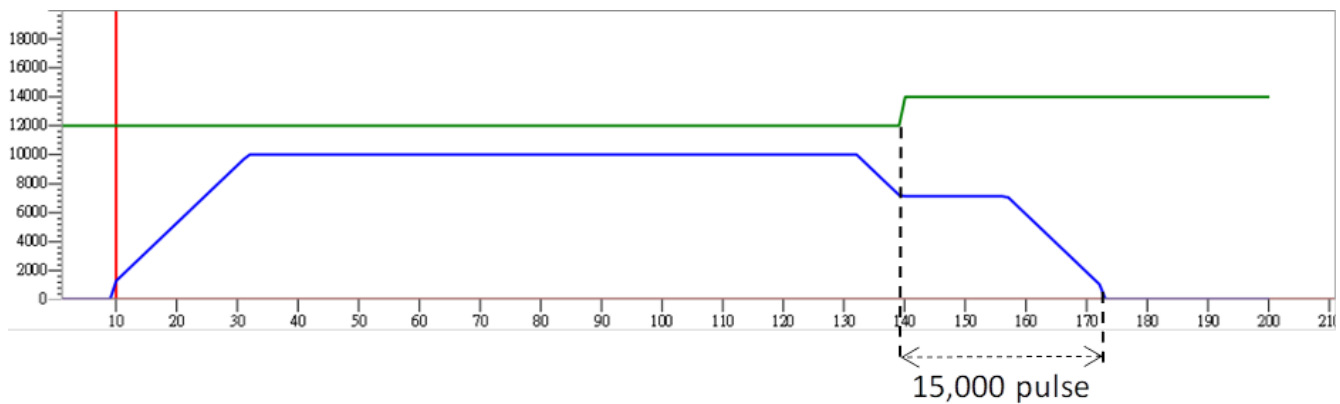


Case 1: when the mark alignment occurs in the ramp-down section

- Green line: signal of mark alignment
- Blue line: the actual output speed

The operation of the mark alignment (green line: OFF -> ON) begins when the number of pulses is 95,000.

Set $S_6 = 15,000$; the output stops after 15,000 pulses at the current speed.



Case 2: when the mark alignment occurs in the ramp-down section

- Green line: signal of mark alignment
- Blue line: the actual output speed

The operation of the mark alignment (green line: OFF -> ON) begins when the number of pulses is 95,000. After the mark alignment, set S_6 to -1, the output stops immediately.

